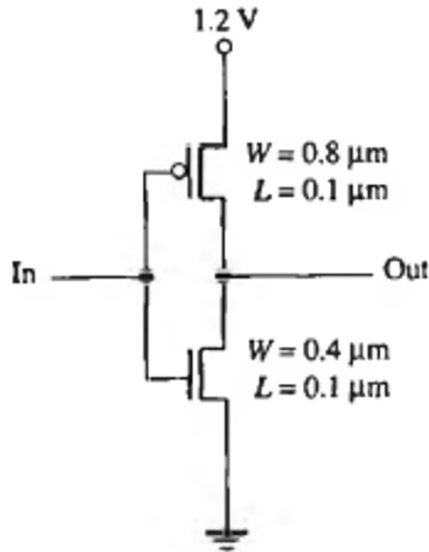


Q1a)



CMOS: No steady state current flow, so $V_{OH} = V_{DD}$, $V_{OL} = GND$

$$V_{OH} = 1.2V, V_{OL} = 0V$$

$$V_{IL} = \frac{2V_{out} - V_{DD} - |V_{TP}| + (k_N/k_P)(V_{TN})}{1 + (k_N/k_P)} \quad V_{out} \approx 1.2V \text{ as first guess}$$

From lecture slide 7-18

$$V_{IL} = \frac{2(1.2) - 1.2 - 0.4 + 1.93(0.4)}{1 + 1.93}$$

$$V_{IL} = 0.537V$$

$$V_{TP} = -0.4V$$

$$V_{TN} = 0.4V$$

$$k_N/k_P = \frac{(W_N/L_N)\mu_N}{(W_P/L_P)\mu_P}$$

$$k_N/k_P = \frac{W_N\mu_N}{W_P\mu_P} = \frac{0.4 \cdot 270}{0.8 \cdot 70}$$

$$k_N/k_P = 1.93$$

At $V_{in} = V_{IL}$ (Region II), $I_{DN}(sat) = I_{DP}(lin)$

$$\frac{W_N\mu_{sat}C_{ox}(V_{in} - V_{TN})^2}{(V_{in} - V_{TN}) + E_{CN}L_N} = \frac{W_P}{L_P} \cdot \frac{\mu_P C_{ox}}{\left(1 + \frac{V_{DD} - V_{out}}{E_{CP}L_P}\right)} \times \left[(V_{DD} - V_{in} - |V_{TP}|)(V_{DD} - V_{out}) - \frac{(V_{DD} - V_{out})^2}{2} \right]$$

From lecture notes slide 7-17

Plug in $V_{in} = V_{IL}$, solve for V_{out}

$$\frac{4 \cdot 10^{-5} (8 \cdot 10^6) (0.537 - 0.4)^2}{(0.537 - 0.4) + 0.6} = 8 \frac{70}{1 + \frac{1.2 - V_{out}}{2.4}} \left[(1.2 - 0.537 - 0.4)(1.2 - V_{out}) - \frac{(1.2 - V_{out})^2}{2} \right]$$

$$V_{out} = 1.14V$$

Solve V_{IL} again with new V_{out}

$$V_{IL} = \frac{2(1.14) - 1.2 - 0.4 + 1.93(0.4)}{1 + 1.93}$$

$V_{IL} = 0.5V$, Solve for V_{out} with new V_{IL}

$$\frac{4 \cdot 10^{-5} (8 \cdot 10^6) (0.5 - 0.4)^2}{(0.5 - 0.4) + 0.6} = 8 \frac{70}{1 + \frac{1.2 - V_{out}}{2.4}} \left[(1.2 - 0.5 - 0.4)(1.2 - V_{out}) - \frac{(1.2 - V_{out})^2}{2} \right]$$

$V_{out} = 1.172V$, repeat solving for V_{IL} , V_{out}

$$V_{IL} = 0.517$$

$V_{out} = 1.16V$ } small change, stop iteration

$$\boxed{V_{IL} = 0.51V}$$

$$V_{IH} = \frac{2V_{out} + V_{TN} + (k_P/k_N)(V_{DD} - |V_{TH}|)}{1 + (k_P/k_N)}$$

From lecture slide 7-20

$V_{out} = 0$ as initial guess

$$\frac{k_P}{k_N} = \frac{\mu_P C_{ox} W_P / L}{\mu_N C_{ox} W_N / L} = \frac{70 \cdot 8}{270 \cdot 4} = 0.519$$

$$V_{IH} = \frac{2(V_{out}) + 0.4 + 0.519(1.2 - 0.4)}{1 + 0.519}$$

$$V_{IH} = \frac{2(V_{out}) + 0.815}{1.519} = \frac{2(0) + 0.815}{1.519}$$

$$V_{IH} = 0.537$$

At $V_{in} = V_{IH}$ (Region IV), $I_{DN}(lin) = I_{DP}(sat)$

$$\frac{W_N}{L_N} \frac{\mu_N C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_N}\right)} \left[(V_{in} - V_{TN}) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_P \mu_{sat} C_{ox} (V_{DD} - V_{in} - |V_{TH}|)^2}{(V_{DD} - V_{in} - |V_{TH}|) + E_{CP} L_P}$$

Solve for V_{out} ,

From lecture slide 7-19

$$4 \frac{270}{1 + \frac{V_{out}}{0.6}} \left[(0.537 - 0.4) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{8 \cdot 10^{-5} \cdot 8 \cdot 10^6 (1.2 - 0.537 - 0.4)^2}{(1.2 - 0.537 - 0.4) + 2.4}$$

$$V_{out} = 0.11 \pm 0.136i \text{ ignore}$$

Resolve for V_{IH} ,

$$V_{IH} = \frac{2(0.11) + 0.815}{1.519}$$

$$V_{IH} = 0.683$$

Resolve for V_{out} ,

$$4 \frac{270}{1 + \frac{V_{out}}{0.6}} \left[(\underline{0.683} - 0.4) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{8 \cdot 10^{-5} \cdot 8 \cdot 10^6 (1.2 - \underline{0.683} - 0.4)^2}{(1.2 - \underline{0.683} - 0.4) + 2.4}$$

$V_{out} = 0.012$, keep iterating

$$V_{IH} = 0.552$$

$$V_{out} = 0.056$$

$$V_{IH} = 0.61$$

Converges to $V_{IH} = 0.59V$, $V_{out} = 0.039V$

$$V_s = \frac{V_{DD} - |V_{TP}| + \chi V_{TN}}{1 + \chi}$$

$$\chi = \sqrt{\frac{\mu_n W_n}{\mu_p W_p}} = \sqrt{\frac{270 \cdot 4}{70 \cdot 8}}$$

$$V_s = \frac{1.2 - 0.4 + 1.39(0.4)}{1 + 1.39}$$

$$\chi = 1.39$$

$$V_s = 0.567V$$

Q1b)

With W_P doubled, V_{OL} and V_{OH} would remain unchanged, but increased pull-up drive strength would increase V_{IL} , V_{IH} , and V_S

1c)

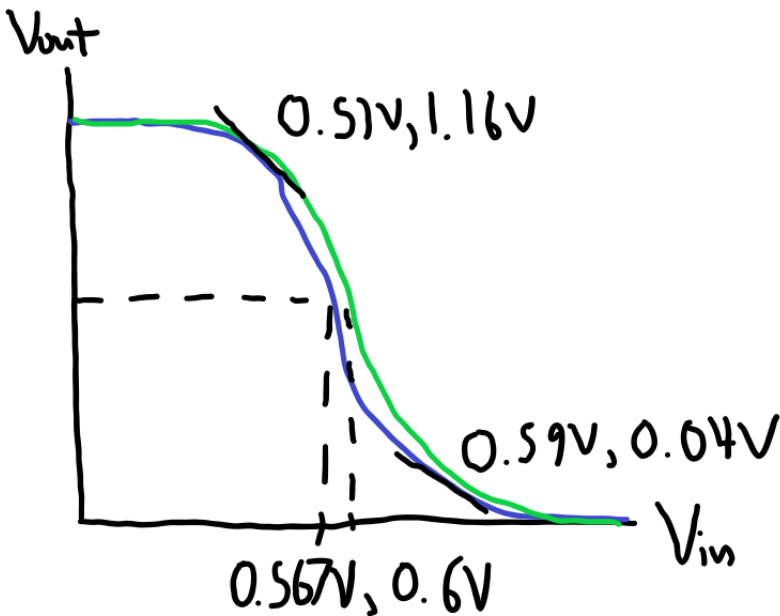
$$V_S = \frac{V_{DD} - |V_{TP}| + \chi V_{TN}}{1 + \chi}, \quad \chi = \sqrt{\frac{\mu_n W_n}{\mu_p W_p}} = \sqrt{\frac{270 \cdot 4}{70 \cdot 16}}$$

$$\chi = 0.98$$

$$V_S = \frac{1.2 - 0.4 + 0.98(0.4)}{1 + 0.98}$$

$$V_S = 0.6V$$

1d)



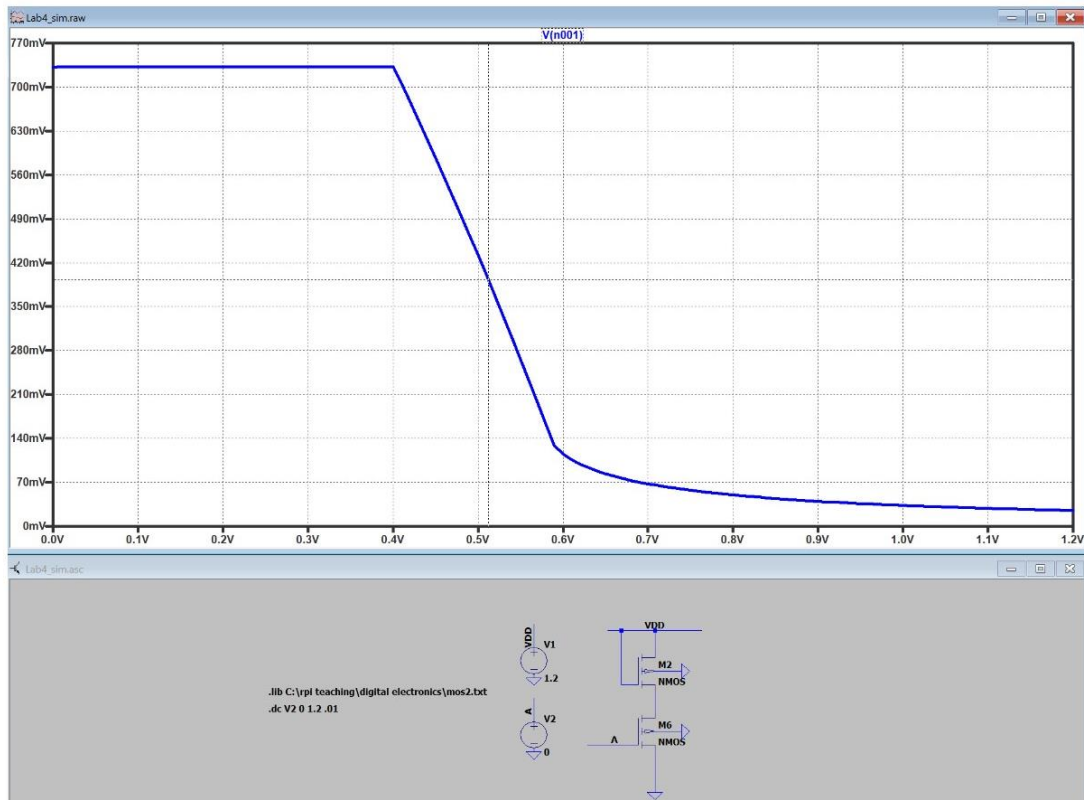
part a

$$V_S = 0.567V$$

$$V_{IL} = 0.51V$$

$$V_{IH} = 0.59V$$

Q2a)



$$\begin{aligned}
 N_{MH} &= V_{OH} - V_{IH}, & V_{OH} &= 0.735V, V_{OL} = 0.025V \text{ by inspection} \\
 N_{MH} &= 0.735 - 0.58 = 0.155V & V_{IL} &= 0.4V, V_{IH} = 0.58V \text{ by inspection} \\
 N_{ML} &= V_{IL} - V_{OL}, \\
 N_{ML} &= 0.4 - 0.025 = 0.375V \\
 V_S &= 0.51V
 \end{aligned}$$

Q2b) Pull-up NMOS has large V_{SB} , and thus body effect.

$$\begin{aligned} \text{At } V_{OH}, \quad V_{GS} - V_T &= 0 & V_{out} &= V_S \\ V_G - V_S - V_T &= 0 & V_{SB} &= V_{out} \end{aligned}$$

$$1.2 - V_{out} - V_T = 0$$

$$V_T = V_{T0} + \gamma (\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F})$$

$$V_T = 0.4 + 0.2 (\sqrt{V_{out} + 0.88} - \sqrt{0.88})$$

$$1.2 - V_{out} = V_T$$

$$1.2 - V_{out} = 0.4 + 0.2 (\sqrt{V_{out} + 0.88} - \sqrt{0.88})$$

$$V_{out} = V_{OH} = \underline{0.734V} \text{ (this is identical to lab 3, Q2)}$$

For V_{OL}

Assuming V_{OL} is low, upper NMOS will be saturated. Lower will be linear

$$\begin{aligned} I_{D\text{upper}} &= I_{D\text{lower}} \\ \text{assuming } V_{OL} \text{ is low also assumes } V_{SB} \text{ is low} \\ W_u \mu_{\text{sat}} C_{ox} \frac{(V_{DD} - V_{out} - V_T)^2}{V_{DD} - V_{out} - V_T + E_c L} (1 + \lambda (V_{DD} - V_{out})) &= \frac{\mu_n C_{ox}}{1 + \frac{V_{out}}{E_c L}} \cdot \frac{W_L}{L} \left[(V_{in} - V_T) V_{out} - \frac{V_{out}^2}{2} \right] (1 + \lambda V_{out}) \\ 10^{-5} \cdot 8 \cdot 10^6 \cdot \frac{(1.2 - V_{out} - 0.4)^2}{1.2 - V_{out} - 0.4 + 0.6} (1 + 0.2(1.2 - V_{out})) &= \frac{270 \cdot 10}{1 + \frac{V_{out}}{0.6}} \left[(1.2 - 0.4) V_{out} - \frac{V_{out}^2}{2} \right] (1 + 0.2 \cdot V_{out}) \end{aligned}$$

$$V_{out} = 0.021V$$

$$\underline{V_{OL} = 0.021V}$$

$$V_{out} \text{ at } V_S = \frac{V_{OH} + V_{OL}}{2} = \frac{0.734 + 0.021}{2} = 0.378 \text{ V}$$

Solve for V_T at $V_{out}(V_S)$,

$$V_T(V_S) = 0.4 + 0.2(\sqrt{0.378 + 0.88} - \sqrt{0.88})$$

$$V_T(V_S) = 0.44 \text{ V}$$

Now we can solve for V_S

$$I_{DN}(\text{sat}) = I_{DP}(\text{sat})$$

Use V_{out} , not V_{in}

$$\frac{W_N \mu_{sat} C_{ox} (V_{in} - V_{TN})^2}{(V_{in} - V_{TN}) + E_{CN} L_N} = \frac{W_P \mu_{sat} C_{ox} (V_{DD} - \boxed{V_{in}} - |V_{TP}|)^2}{(V_{DD} - \boxed{V_{in}} - |V_{TP}|) + E_{CP} L_P}$$

$$\frac{10(V_{in} - 0.4)^2}{(V_{in} - 0.4) + 0.6} = \frac{1(1.2 - V_{out} - 0.44)^2}{1.2 - V_{in} - 0.44 + 0.6}$$

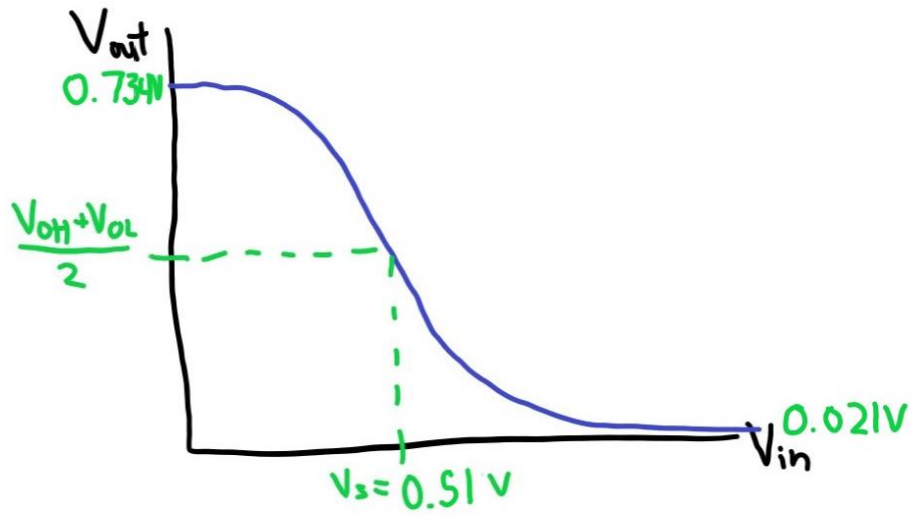
$$\frac{10(V_{in} - 0.4)^2}{V_{in} + 0.2} = \frac{(0.76 - V_{out})^2}{1.36 - V_{in}}$$

$$\frac{10(V_{in} - 0.4)^2}{V_{in} + 0.2} = \frac{(0.76 - 0.378)^2}{1.36 - V_{in}}, \quad V_{in} = V_S$$

$$\boxed{V_S = 0.51 \text{ V}}$$

$$\text{Use of } V_S = \frac{V_{DD} - |V_{TP}| + \chi V_{TN}}{1 + \chi} \text{ yields } V_S = 0.44 \text{ V}$$

and is also accepted, despite it not quite being correct



VTC plots are identical, outside neglecting to calculate V_{IL} and V_{IH} for more accurate curvature.